

Practitioner Interview Protocol

Replication Package for "When Parsing Goes Wrong: Revisiting Log Anomaly Detection from Parsing to Deployment"

1. Purpose

This protocol describes the practitioner interviews used to complement the quantitative experiments in the study. The interviews examined how engineers understand industrial logs, how they perceive parsing errors and anomaly detection, and how they evaluate the use of external public logs in data-scarce industrial settings.

2. Participants

The study interviewed 11 practitioners who were directly involved in the development, operation, maintenance, or reliability assurance of the four industrial systems. Participants had practical experience with system logs, debugging, monitoring, incident response, or anomaly detection. Their roles covered software development, system operation, reliability engineering, and system maintenance.

3. Interview Format and Procedure

The interviews followed a semi-structured format. Before each interview, participants received a concise summary of the study goals and main empirical findings, including the identified parsing-error types, the observed impact of parsing quality on anomaly detection, and the mixed effect of using external public logs. Each interview lasted approximately 45 to 60 minutes and was conducted online or in person. Interviews were recorded and transcribed with participant consent.

At the beginning of each interview, the interviewer explained the purpose of the study, the voluntary nature of participation, confidentiality requirements, and the anonymization procedure. Participants were instructed not to disclose sensitive operational identifiers, proprietary system details, user records, credentials, or confidential business information.

4. Interview Questions Used in the Study

The following 28 questions were used to guide the semi-structured interviews. Follow-up questions were asked only to clarify participant responses or request concrete examples.

A. Participant Background

1. What is your current role, and what are your main responsibilities in the industrial system?
2. How many years of experience do you have in software development, system operation, reliability engineering, or related work?
3. How do you use logs in daily work, such as debugging, monitoring, incident response, or reliability analysis?

B. Industrial Log Characteristics and Public Benchmarks

4. How would you describe the logs generated by your system in terms of structure, length, and domain-specific information?
5. Which characteristics make these logs difficult to parse, understand, or analyze automatically?
6. Do different modules, devices, services, or development teams follow consistent logging styles?
7. How often do log formats change because of software updates, configuration changes, operational changes, or hardware changes?

8. Compared with public benchmark logs, what characteristics of industrial logs are more complex or more challenging?

C. Parsing Errors and Practical Consequences

9. Have you observed cases where log parsing tools produced incorrect or misleading templates?
10. How do template over-merging and template over-splitting affect debugging, monitoring, or anomaly detection?
11. Which variables or parameters must be preserved accurately in your system logs?
12. How do new software versions, rare events, or evolving log formats create parsing problems?
13. Is semantic rewriting by an LLM-based parser acceptable when the rewritten template no longer faithfully preserves the raw log message?

D. Anomaly Detection in Practice

14. How are anomalies currently detected in your system?
15. Which types of anomalies are most important to detect early?
16. What are the operational consequences of false positives and false negatives?
17. What information must an anomaly detection tool provide to make its output actionable for engineers?

E. Data Scarcity and Labeling Constraints

18. Is it difficult to obtain labeled anomalous logs in your system?
19. Who usually confirms whether a log event or incident is anomalous, and what makes this process costly?
20. How do rare normal events and limited labeled anomaly data affect model training, evaluation, and the choice between supervised and semi-supervised anomaly detection methods?

F. External Public Logs and Cross-Dataset Augmentation

21. Do you think public logs can help improve anomaly detection in industrial systems with limited labeled data?
22. Under what conditions are external logs useful for industrial anomaly detection?
23. Under what conditions do external logs become misleading or harmful?
24. Which kinds of similarity between public logs and industrial logs matter most for avoiding semantic mismatch and negative transfer: lexical similarity, template similarity, domain meaning, or operational context?

G. Deployment Requirements and Reflection

25. What requirements must a log parsing or anomaly detection tool satisfy before deployment?
26. How important are robustness, faithfulness, traceability, and human review in real-world deployment?
27. Do the empirical findings of this study match your experience with industrial logs and anomaly detection?
28. What practical challenges in industrial log analysis should future research address?

5. Post-Interview Processing

- Audio recordings were stored securely after participant consent was obtained.
- Interview recordings were transcribed and checked for transcription errors.

- Participant names, company names, internal system names, hostnames, IP addresses, user identifiers, project names, proprietary event codes, and other sensitive details were removed from the transcripts.
- Each participant was assigned an anonymous identifier, such as P1, P2, ..., P11.

6. Thematic Analysis

The interview transcripts were analyzed using thematic analysis. Two researchers independently read the transcripts, generated initial codes, compared and refined the codebook, and grouped codes into higher-level themes. The analysis focused on industrial log characteristics, differences from public benchmarks, parsing-error consequences, data scarcity, external-log augmentation, negative transfer, model trust, and deployment requirements.

Divergent views were retained when they provided meaningful contrast. The resulting themes were mapped back to the quantitative findings to identify whether they supported, explained, qualified, or extended the experimental results. Thematic saturation was assessed during coding. After the ninth interview, no new major themes emerged, and the final interviews mainly confirmed or refined existing themes.

7. Member Checking and Reporting

After theme development, participants received concise anonymized summaries of the interpreted themes for confirmation. Reported findings used participant identifiers rather than names, avoided sensitive industrial information, and presented interview results as complementary qualitative evidence rather than statistically representative claims.

8. Confidentiality Requirements

The public replication package excludes real participant names, company names, internal system names, internal hostnames, IP addresses, user identifiers, proprietary event codes, sensitive operational failures, security-related vulnerabilities, confidential business workflows, and any information restricted by industrial agreements.

9. Expected Outputs

- An anonymized participant summary table
- Anonymized interview transcripts
- A refined qualitative codebook
- A theme summary document
- A mapping between qualitative themes and quantitative findings
- Member-checking records
- A final anonymized protocol suitable for inclusion in the replication package

10. Quality-Control Checklist

- All participants provided informed consent.
- All transcripts were anonymized before analysis.
- Two researchers independently coded the transcripts.
- Coding disagreements were discussed and resolved.
- Divergent views were retained when meaningful.
- Thematic saturation was assessed.
- Member checking was conducted.
- No sensitive industrial information is included in public materials.